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Rotaxanes—novel photonic molecules

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Abstract

Rotaxanes are a new class of interesting macromolecules, composed from two interlocked, mobile parts. They can be processed into good optical quality thin films and exhibit interesting second and third-order nonlinear optical properties. In solution the macrocycle rotate around the thread, as it was shown previously by the electro-optic Kerr effect and NMR. The measured frequency of rotation agrees well with theoretical calculations and depends on temperature as well as on the applied external field strength. Rotaxanes can be poled, similarly as functionalized polymers. Using the third harmonic generation technique a clipping movement was demonstrated when shining in the thread absorption band.
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1. Introduction

At present there is a great interest in the design and synthesis of new, addressable, functional organic molecules for use in various types of practical applications. Mechanically interlocked hydrogen bond assembled rotaxanes (cf. Fig. 1) are organic systems which offer unique architectural and structural properties [1,2]. They are a class of mechanically interlocked molecules where a macrocyclic ring is locked onto a thread by two bulky stoppers [3] (cf. Fig. 1). They have attracted great attention as promising candidates for the development of prototypical structural units for device applications because macrocycles can rotate and translate back and forward along the chain. Rotaxane architectures are thus particularly attractive because the components of the molecule are held together by a dynamic mechanical bond which can be controlled at the molecular level by applying an external stimuli [4]. For these reasons they have been proposed as nanoscale devices such as switchable molecular brakes [5], shuttles [6], ratches [7] and electronically configurable logic gates [8]. Moreover the absence of chromophores or redox sensitive groups make rotaxanes an ideal backbone onto which additional electro-optically active subunits can be assembled to form a new type of interlocked artificial functional materials for useful electronics or optical properties.

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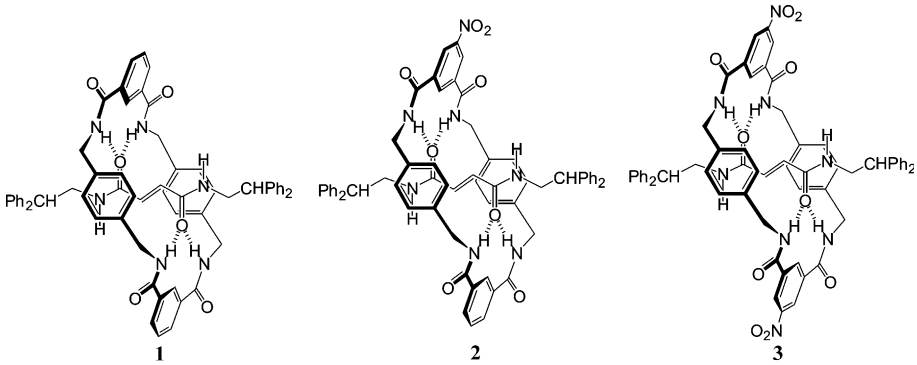

Fig. 1. Chemical structure of studied rotaxanes.

Dynamics of the mechanical bonds linking the ring, the thread and the stoppers components of the rotaxane compounds occupy a special place due to the specific degrees of freedom they have. Indeed, rotaxanes exhibit a large rotational mobility and can be poled by the applied electric field, similarly as functionalized polymers.

These degrees of freedom, not available in other materials, can be used to switch the molecular properties through the adequate changes in the local environment, chemical stimuli and external optical or electrical applied fields. Indeed, using the electro-optic Kerr effect we have shown that the macrocycle rotates around the thread with frequency depending on the strength of external electric field and on the temperature [9]. These results are confirmed by independent NMR measurements and by theoretical modelling [9].

In this paper we describe linear and nonlinear optical properties of a serie of rotaxanes. In particular we show that rotaxanes can be processed into good optical quality thin films by vacuum evaporation. The linear optical properties of thin films, obtained by vacuum evaporation can be controlled by an appropriate functionalization of macrocycle. The films can be poled, with poling and depoling kinetics very similar to that of functionalized polymers. In solution the macrocycle rotates, as we have shown it by electro-optic Kerr effect and confirmed by NMR measurements. The speed of rotation can be controlled with applied electric field.

2. Molecule synthesis and thin film preparation

Rotaxanes were synthesized by the "clipping" methodology, consisting on the simultaneous slow addition of solutions of isophthaloyl dichloride and *p*-xylylene diamine in chlorinated solvents in the presence of the respective threads, with triethylamine as base. The thread provides the template information to form the benzylic amide macrocycle around itself via intermolecular hydrogen bond interactions (efficiency was more than 97% in the formation reaction, indicating close complementarity between the macrocycle and the thread). The three rotaxanes have the same fumaric thread while macrocycle differs from pure fumrot **1** in the addition of one nitro group in the case of mono-nitro fumrot **2** and two nitro groups in the case of di-nitro fumrot **3** as shown in Fig. 1.

Thin films of each rotaxane were deposited on fused silica and glass slides by vacuum sublimation. The vacuum level during deposition was 10^{-6} T, the powder and target temperatures were 220 and 25 °C respectively, and the deposition rate was varied from 100 Å/s at the beginning of the procedure to 10–15 Å/s after the first 100 nm thickness was obtained and then until the end of the process.

3. Linear optical properties

The refractive indices of rotaxane thin films were measured in the visible and near infrared, at

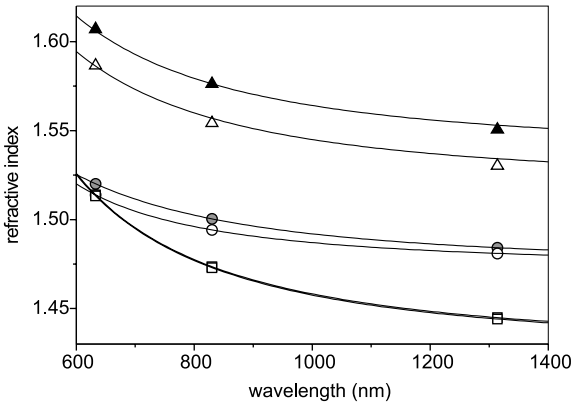

Fig. 2. Refractive index dispersions in thin films of rotaxanes (Δ—1, ◯—2, □—3) shown in Fig. 1. Open marks denote TM while closed the TE refractive index. Continous lines depict fitted values using the Sellmeier equation.

623.8, 830 and 1314 nm, by means of m-lines spectroscopy. This technique allows to measure both the refractive index and the thickness when the films support more than one guided mode. It allows also to measure the optical birefringence by choosing the adequate polarization of the coupled light into thin film. Fig. 2 shows the measured and fitted with Sellmeier equation the wavelength dependence of refractive index of thin films of rotaxanes **1–3** (cf. Fig. 1) for both TE and TM polarizations. A large optical birefringence is observed in thin films of pure fumrot (**1**). The attachment of the nitro group to macrocycle results in a decrease of the order in the case of mono substituted rotaxanes and its disappearance for the di-substituted one. In the last case the films are almost isotropic. This is a clear probe of the influence of the nitro group on the growth of thin films and on the amount of order.

4. Poling

The films were poled using the experimental setup described elsewhere [10]. The amount of polar order was measured by in situ second harmonic generation. Fig. 3 shows the build up of SHG signal as function of poling time and at different temperatures. It is seen that this behavior is

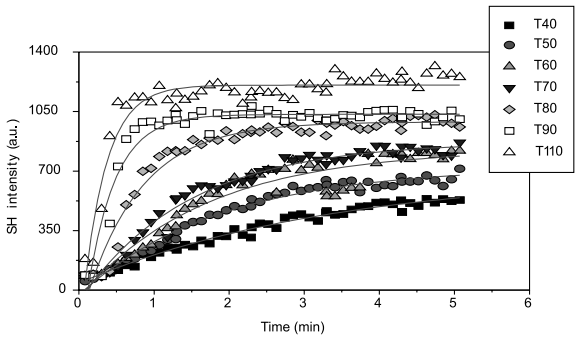

Fig. 3. Temporal variation of SH intensity with poling time at different temperatures for pure fumrot **1**.

very similar to what is observed in functionalized polymers. With increasing temperature the mobility is increasing and the polar ordering time constant is decreasing. There is an optimum poling temperature, corresponding to ≈110 °C at which the poling is most efficient. Calibration with SHG measurements on a *y*-cut quartz single crystal ($d = 0.5$ pm/V), done at the same conditions, yields $d_{33} = 3.4 \pm 0.3$ pm/V. This value is relatively small but should be surely increased by an appropriate functionalization of rotaxanes. The SHG measurements were done on pure fumrot only and their main aim was the demonstration of the ability of rotaxanes to be poled by a static electric field. We expect similar results with other rotaxanes from this family (cf. Fig. 1).

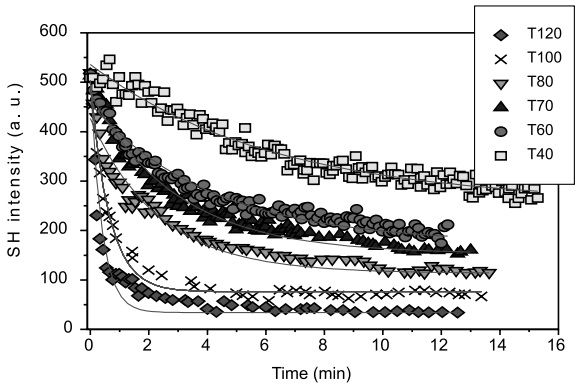

Fig. 4. Decay of SH intensity vs. time at different temperatures for pure fumrot **1**.

Similarly, the decay of the polar order is very similar to that observed in functionalized polymers, as it is seen from Fig. 4 showing SHG intensity variation at different temperatures after switching of the poling field. The data can be well fitted by a single exponential function (solid lines in Fig. 4), except the two highest temperatures, where more complex dependence is observed, from which the relaxation time can be estimated. Again the relaxation time depends on temperature, decreasing with its increase. In particular the estimated in this way relaxation times at 40 and 120 °C are 375 and 23.8 s, respectively.

5. Third-order NLO properties

5.1. Quadratic electro-optic Kerr effect mesurements

Using the experimental setup described elsewhere [11] we have studied the frequency dependence of the quadratic electro-optic Kerr constant in solution in an apolar solvent-dioxane and at room temperature. Fig. 5 shows the frequency variation of the Kerr constant for pure fumrot at different applied electric fields. First of all a resonance enhancement is observed, whose position depends on the applied electric field. When increasing the field strength the resonance frequency shifts towards lower values with a simultaneous broadening of the resonance peaks. At low electric field a shoulder is observed at slightly higher frequency which is disappearing with increasing strength of electric field. The regime time at which resonance peaks appear generally corresponds with the large scale intramolecular motion, which can be probed by nuclear magnetic resonance (NMR) experiments. The dynamic behavior of rotaxane **1**, measured by variable temperature (VT) ^{1}H NMR spectroscopy, has been already published [9]. It has been shown that there exists a rate of macrocyclic ring rotation about the thread of 340 Hz at zero field and a barrier of 13.4 ± 0.1 k/cal/mol at 253 K and it is also observed that other intramolecular motions are occurring on a similar time scale to macrocycle circumrotation about the thread, which will correspond with the little shoulder observed at low applied electric field. The observed increase of the resonance frequency with decreasing electric field in electro-optic Kerre effect experiments corroborate well with the observed higher rotation frequency by NMR at zero field.

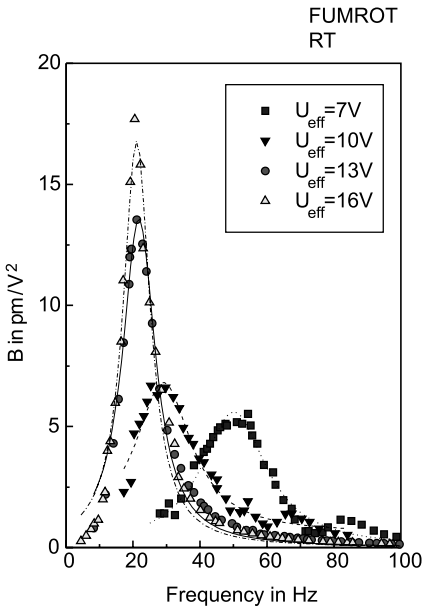

Fig. 5. Variation of the quadratic electro-optic Kerr constant in function of the frequency of the applied electric field and for its different values.

To give a more detailed understanding of the dynamics involved in both rotaxane systems, the submolecular motions were simulated through modeling for related benzylic amide catenane systems using the MM3 potential and the TINKER program as it is shown in Ref. [9]. Rotaxane **1** presents a more complicated picture in which the pirouetting is coupled to pivoting, or scissoring, of the macrocycle against the thread, with the ideal fulcrum of the motion located near the centre of mass of the molecule. The calculated activation energy is 13.9 k/cal/mol, which compare well with that determined by NMR experiments.

5.2. Third harmonic generation experiments

The THG measurements were performed on vacuum evaporated thin films using the experimental setup and the methodology described else-

Table 1
Third-order NLO susceptibilities of studied rotaxanes as measured by THG technique at 1907 nm fundamental wavelength and calibrated with silica with susceptibility of $\chi^{(3)}(-3\omega;\omega,\omega,\omega) = 2.8 \times 10^{-14}$ esu [13]

Rotaxane	Fumrot (**1**)	Mono-nitro fumrot (**2**)	Di-nitro fumrot (**3**)
$\chi^{(3)}(-3\omega;\omega,\omega,\omega)$ (esu)	$1.809(\pm 0.006) \times 10^{-12}$	$1.834(\pm 0.004) \times 10^{-12}$	$1.990(\pm 0.007) \times 10^{-12}$

where [12]. They were done at 1907 nm fundamental wavelength, thus far from the absorption band for both fundamental and harmonic waves. All measurements were done in vacuum in order to avoid the air contribution. The $\chi^{(3)}(-3\omega;\omega,\omega,\omega)$ susceptibility were obtained by calibration with THG measurements performed on a silica plate at the same experimental conditions. For silica we used the usually admitted value of $\chi^{(3)}(-3\omega;\omega,\omega,\omega) = 2.8 \times 10^{-14}$ esu, as determined by Meredith et al. [13]. It is very easy to recalibrate these values to another values of the used standard. The cubic susceptibility values determined in this way are listed in Table 1. They are very close to similar systems with a similar π electron conjugation [12]. In fact, these values depend strongly on the electron delocalization in organic molecules. We observe a monotonic and small decrease of $\chi^{(3)}(-3\omega;\omega,\omega,\omega)$ susceptibility when passing from pure fumrot to Di-nitro fumrot (3). Similar behavior is observed for refractive index and may just correspond to a worsening packing.

Very interesting behavior of THG susceptibility is observed when shining thin film in the C=C absorption band (around 350 nm) as it is seen in Fig. 6. A significant decrease of $\chi^{(3)}(-3\omega;\omega,\omega,\omega)$ susceptibility is observed due to the illumination. When this illumination is switched off, the $\chi^{(3)}$ susceptibility recovers very slowly its initial after (3–4 days). We believe that this is due to the *trans–cis* izomerization process. Indeed in *cis* form, due to the smaller π electron conjugation the $\chi^{(3)}(-3\omega;\omega,\omega,\omega)$ is smaller too. Usually the transitions, from *trans* to *cis* form are much faster then the inverse ones. This is due to the fact that the last transitions go through nonradiative channels. Also in solid the izomerization process is much slower than in solution, and sometimes it is even blocked.

The reversible *trans–cis* izomerization was observed also by checking variation of the absorption spectrum due to the UV illumination. The 350 nm band shifts to the lower wavelength. After the illumination stops it recovers its initial shape [14]. However here we show that using NLO techniques we can also follow the photoizomerization processes.

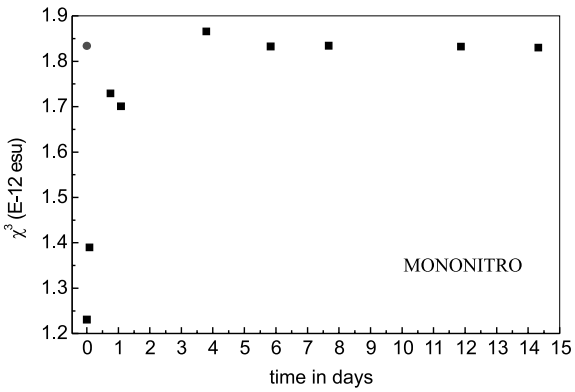

Fig. 6. Variation of THG susceptibility of mono-nitro fumrot under illumination in the C=C absorption band. Closed circle shows the initial value whereas squares those after illumination and as a function of time.

6. Conclusions

Rotaxanes appear to be interesting, functional materials, whose properties can be controlled by external stimuli, like electric field and light. The order of vacuum deposited films depends on the functionalization of macrocycle. Pure fumrot gives anisotropic films whereas di-substituted with NO_2 group forms isotropic ones.

The films can be poled with a static external field. Rotaxanes behave very similar to functionalized polymers. Although the measured value of d coefficient for fumrot is relatively low ($d_{33} = 3.4 \pm 0.3$ pm/V), it could be increased by an appropriate functionalization of this molecule, as already mentioned.

The macrocycle rotates around the thread with velocity depending on its chemical structure. The speed of rotation can be controlled by the applied electric field. It decreases with its increasing strength.

Very recently the shuttling of macrocycle was also demonstrated [15]. The photoizomerizable thread (cf. Fig. 1) offers also the opportunity to realize a clipping movement which is of great interest for practical applications, such as trapping and relaxing of atoms, optical memories, etc. This was evidenced by both: linear absorption measurements and third order susceptibility variation under irradiation in the thread absorption band.

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